

TRANSPORT MANAGEMENT SYSTEM (TMS)

PROJECT PROPOSAL DOCUMENT – ADVANCED VERSION

1. PROPOSAL SUMMARY

We propose to develop an **Advanced Transport Management System (TMS)** with real-time GPS live tracking, AI-based route optimization, customer and driver mobile apps, automated billing, and comprehensive fleet analytics. Ideal for **large transport companies and logistics operators (80-300 vehicles)** seeking full digital automation, better fleet visibility, and cost optimization.

Ideal For: Large fleet operators, logistics companies, city cab aggregators, corporate transport providers (80-300 vehicles)

2. OBJECTIVES

- Achieve real-time visibility across all vehicles and trips
- Automate booking, dispatch, billing, and driver communication
- Optimize routes and reduce fuel and operational costs with AI
- Improve customer experience with live tracking and notifications
- Enable data-driven decision making with advanced analytics

3. SCOPE OF WORK

3.1 Website + Customer Booking Portal (Advanced)

- Premium transport website with modern UI/UX
- Online cab / truck / bus booking portal
- Real-time vehicle availability and fare estimate
- Customer account portal (booking history, invoices)
- Corporate client self-service booking portal
- Live shipment tracking page for customers

Cost: Rs. 42,000 - Rs. 62,000

3.2 Real-Time GPS Live Tracking

- Live vehicle location on map dashboard
- Geo-fence alerts (entry/exit of defined zones)
- Idle time, over-speed and stoppage alerts
- Historical trip route playback
- Customer-facing live tracking link per trip

Cost: Rs. 55,000 - Rs. 80,000

3.3 AI Route Optimization & Dispatch

- AI-based shortest and fastest route suggestion
- Smart vehicle-to-trip auto assignment
- Multi-stop delivery route planning
- Traffic-aware dynamic re-routing
- Load optimization for goods transport

Cost: Rs. 55,000 - Rs. 80,000

3.4 Advanced Fleet Management

- Complete vehicle lifecycle management
- Predictive maintenance alerts based on km / usage
- Tyre, battery and parts replacement tracking
- Multi-depot fleet management
- Fleet cost per km and utilization analysis

Cost: Rs. 45,000 - Rs. 65,000

3.5 Driver Management + Attendance + Safety

- Detailed driver profiles and document management
- Driver attendance and duty hours tracking
- Driving behaviour analytics (speed, braking, idling)
- Driver performance scoring and rating
- Trip allowance and salary management

Cost: Rs. 40,000 - Rs. 60,000

3.6 Freight & Load Management (Goods Transport)

- Consignment creation and tracking
- LR (Lorry Receipt) / E-way bill generation
- Sender and receiver management
- Delivery confirmation with digital POD (Proof of Delivery)
- Freight billing and payment tracking

Cost: Rs. 45,000 - Rs. 65,000

3.7 Billing, Invoicing & Finance

- Auto invoice generation per trip/consignment
- GST-compliant billing
- Corporate client account management
- Payment received and outstanding tracking
- P&L; reports per vehicle, driver and route

Cost: Rs. 35,000 - Rs. 52,000

3.8 Admin Dashboard (Advanced Analytics)

- Real-time fleet and operations dashboard
- Route-wise profitability reports
- Driver and vehicle performance analytics
- Fuel efficiency and cost per trip reports
- Custom report export (PDF/Excel)

Cost: Rs. 35,000 - Rs. 50,000

4. ADDITIONAL FEATURES

4.1 WhatsApp + SMS + Email Notifications

- Booking confirmation and trip alerts
- Live tracking link sharing via WhatsApp
- Document expiry and service due alerts
- Delivery confirmation to customers

Cost: Rs. 18,000 - Rs. 28,000

4.2 E-Way Bill & Compliance Integration

- GST E-way bill generation from TMS
- VAHAN API integration for vehicle verification
- Permit and fitness certificate management

Cost: Rs. 22,000 - Rs. 35,000

4.3 Role-Based Access & Security

- Multi-level roles (Admin/Dispatcher/Driver/Client)
- Secure JWT authentication
- Activity and trip audit logs

Cost: Rs. 15,000 - Rs. 22,000

4.4 Cloud Backup & Data Security

- Daily automated encrypted cloud backup
- Secure data storage and recovery

Cost: Rs. 12,000 - Rs. 18,000

4.5 Fuel Card & Vendor Integration

- Fuel card transaction import and reconciliation
- Vendor / garage / tyre shop management
- Purchase order for maintenance and parts

Cost: Rs. 18,000 - Rs. 28,000

5. TECHNOLOGY STACK (MERN + AI)

- **Frontend:** React.js
- **Backend:** Node.js (Express.js)
- **Database:** MongoDB + Redis
- **GPS:** Google Maps Platform / HERE Maps / Custom GPS device API
- **AI/ML:** Route optimization engine
- **Payment:** Razorpay / PayU
- **Hosting:** AWS Cloud

6. TOTAL PROJECT COSTING

Development Cost (One-time)	Rs. 4,50,000 - Rs. 6,50,000 (One-time)
Infrastructure	Hosting: Rs. 4,000 - Rs. 10,000/month
Infrastructure	Domain + SSL: Rs. 2,000/year
Infrastructure	GPS & Maps API: Rs. 3,000 - Rs. 8,000/month
Maintenance Cost	Rs. 25,000 - Rs. 38,000/month/month

7. PROJECT TIMELINE

Phase	Duration
Planning	1 Week
Development	8-10 Weeks
Testing	1-2 Weeks
Deployment	1 Week
Total Duration	10-13 Weeks

8. LIMITATIONS

- No dedicated mobile apps (can be added separately)
- No IoT-based vehicle diagnostics (OBD)
- No full FASTag / toll integration
- No AI-based demand forecasting for fleet planning

9. FUTURE UPGRADES

- Customer and driver mobile apps
- OBD-based real-time vehicle diagnostics
- FASTag and toll auto-expense integration
- AI fleet demand and capacity forecasting

10. SUPPORT AND WARRANTY

- 1 month free support
- Bug fixing included
- Staff training provided

11. TERMS AND CONDITIONS

- 50% advance payment
- 50% after completion
- Additional features charged separately

12. CONCLUSION

This Advanced TMS delivers a **powerful, real-time, and AI-assisted transport management platform** — with live GPS tracking, smart dispatch, route optimization, e-way bill generation, freight management, and deep analytics, making it the ideal digital backbone for large fleet and logistics operations.

13. CONTACT DETAILS

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